

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 6+ years shipping production web applications end to end — React and TypeScript front ends on Node.js APIs and data layers. Led the zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP at MTailor, reporting to the CTO, and built the real-time sync pipeline that kept two datastores serving live traffic throughout. Comfortable owning data-heavy product surfaces front to back: a real-time stock-price PWA over 2M+ records (Dr Wealth), and 3D product visualisation and Express Shipping features measured in conversion and revenue (MTailor). Polyglot backend — Node.js and Python in production, intermediate Go — with SQL and NoSQL data modelling, Docker/Kubernetes microservices, and production OpenAI/Google Vision integrations. Based in Kampala (UTC+3); 4+ years fully remote with US and Singapore teams.

CORE COMPETENCIES

Full-Stack Web Applications (React, TypeScript, Node.js)

End-to-End Feature Ownership

API & Data Layer Design

Data-Heavy Product Interfaces

Performance as a Feature

SQL & NoSQL Data Modelling

Distributed & Microservice Systems

Docker & Kubernetes

WORK EXPERIENCE

MTailorSeptember 2022 – Present

Software Engineer · US Remote

- Led the end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, owning the technical approach front to back and reporting directly to the CTO.
- Built a 3D product visualisation feature for customers using video overlay and ffmpeg — a data-dense, customer-facing interface that increased buyer conversion.
- Shipped Express Shipping end to end (backend, pricing, ops handoff), generating an additional \$40 revenue per order.
- Designed and implemented the real-time two-way sync between MongoDB and Firestore (Node.js + Google Pub/Sub) — the data and API layer that let both stores serve live production traffic during the cutover.
- Migrated a Webflow marketing site from a Parse backend to Firebase, front end to data layer.
- Consolidated infrastructure off AWS (S3 → GCS via Python tooling), saving the company \$5,000/month.

Node.js · React · JavaScript · Firebase · GCP Pub/Sub · Python · ffmpeg · AWS S3/EC2/EBS · Webflow

MTailorMarch 2022 – July 2022

Contractor (Backend Migration) · US Remote

- Hired via Upwork onto an unfamiliar Firebase/GCP stack; built the MongoDB → Firestore migration scripts with complex processing logic and shipped under contract scrutiny.
- Wrote the Firebase SDK documentation (Node.js and Python) that fellow engineers onboarded from; trained the Ops team on the new dashboard and backup procedures.

Node.js · Firebase · Google Cloud Platform · AWS · JavaScript

Dr WealthAugust 2021 – January 2022

Software Engineer · Singapore Remote

- Led optimisation of a PWA delivering real-time stock market prices — a data-heavy product surface where performance was the feature.
- Built batched jobs querying 2M+ Firestore records to keep customer prices current from Morningstar APIs, optimising query shape over infrastructure.
- Added responsive UI pages to the PWA in React, Tailwind and HTML; extended the backend API layer with Firebase Cloud Functions and Express.js on Heroku.

React · Tailwind · Node.js · Express.js · Firebase · JavaScript

Mind2matter.coJanuary 2022 – July 2022

Software Engineer (Full Stack) · US Remote

- Delivered pixel-accurate React UIs from Figma designs under tight agency timelines, partnering directly with design and reporting to the CTO.
- Built Node.js backends for DeFi applications using Web3.

React · Node.js · Web3 · Firebase · Tailwind · JavaScript

Software Engineer / Team Lead · Uganda Remote

- Led a team of 4 developers building legal-tech systems for NGOs (FIDA Uganda, LASPNET) — code review, architecture decisions and delivery.
- Architected a **distributed microservices backend** for FIDA Uganda's case management app using Apache Kafka, Docker and Kubernetes.
- Built a case management system and a paralegal database; shipped a cross-platform React Native/Expo mobile app with Firebase push notifications.
- Built a USSD service for legal-aid providers on a versioned gRPC contract to a Node.js backend.

Node.js · React Native · Apache Kafka · Docker · Kubernetes · NATS Streaming · gRPC · Firebase

PROJECTS

Real-time Pub/Sub Sync Pipeline MTailor

Live, bidirectional MongoDB ↔ Firestore sync with idempotent message processing and conflict handling — supported parallel production traffic on two datastores for the full duration of a 20+ application migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

Healthcare WhatsApp Chatbot AI in production

Conversational assistant integrating OpenAI with Google Vision for prescription OCR over the WhatsApp Business API — dialogue state, media ingestion and third-party platform limits handled end to end.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for matching users with service providers from a messy, user-generated catalogue.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — **Makerere University, Kampala, Uganda**

2018 – 2022

LANGUAGES

English (fluent, working language for 4+ years with US and Singapore teams) · Kiswahili

SKILLS

Proficient: Node.js, JavaScript, TypeScript, React, HTML, CSS/Tailwind, Firebase, GCP, Python, MongoDB, SQL, Git, Linux

Intermediate: Go, AWS, Rust, React Native

Infrastructure & messaging: Docker, Kubernetes, Apache Kafka, Google Pub/Sub, NATS Streaming, gRPC, ffmpeg